

SENIOR THESIS

Collin R. Guedel

A SENIOR THESIS
PRESENTED TO THE FACULTY
OF PRINCETON UNIVERSITY
IN CANDIDACY FOR THE DEGREE
OF BACHELOR OF ARTS

RECOMMENDED FOR ACCEPTANCE
BY THE DEPARTMENT OF
GEOSCIENCES
Advisor: Professor Xinning Zhang

02 April, 2026

This paper represents my own work in accordance with University regulations,

Collin R. Guedel

Abstract

Human land use, primarily through farming and its resulting land disturbances impacts the soil sink of atmospheric hydrogen maintained by soil microbial communities; this thesis' aim is to assess, both through measurements of soil H₂ uptake rate and bacterial genomics, what governs this rate. Based upon preliminary agricultural data that I assess this summer, it appears that tilling has a significant impact on agricultural soils' uptake ; I would like to study this further by comparing farming practices and the resultant microbial behaviors in comparison to undisturbed, non-agriculture soils.

Acknowledgements

Thanks Mom and Dad!

Contents

Abstract	iii
Acknowledgements	iv
List of Figures	vii
List of Tables	1
Main Text	1
Introduction	1
Methods	1
Results	2
Summer Data	2
Soy - Undisturbed - Long	6
Soy - Disturbed Center - Long	12
Good Soy Runs	16
Fairgrown Farms	18
Kinetic Rate Data Summary	24
Discussion	27
Conclusion	27
Appendices	28

Appendices	29
Supplementary Information	29
Code Appendix	29
References	30

List of Figures

List of Tables

.

Introduction

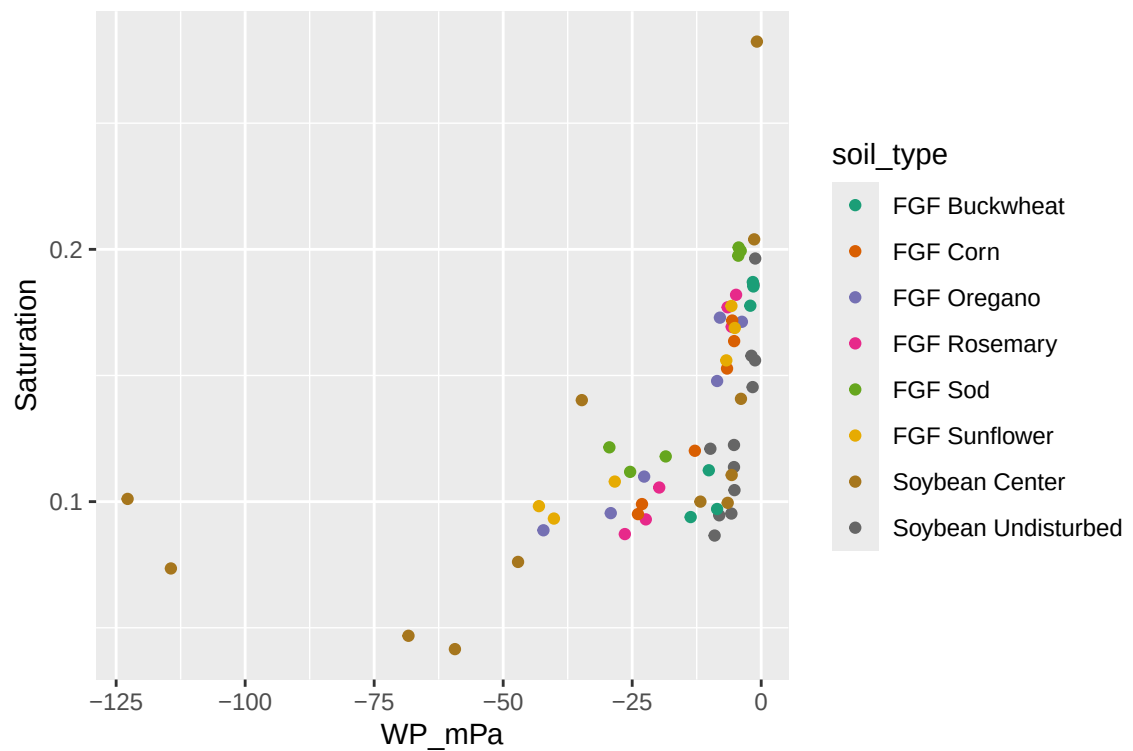
TBD Sand et al. (2023) says

Methods

$$[H_2] = [H_2]_{eq} + ([H_2]_0 - [H_2]_{eq})e^{-kt}$$

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
```

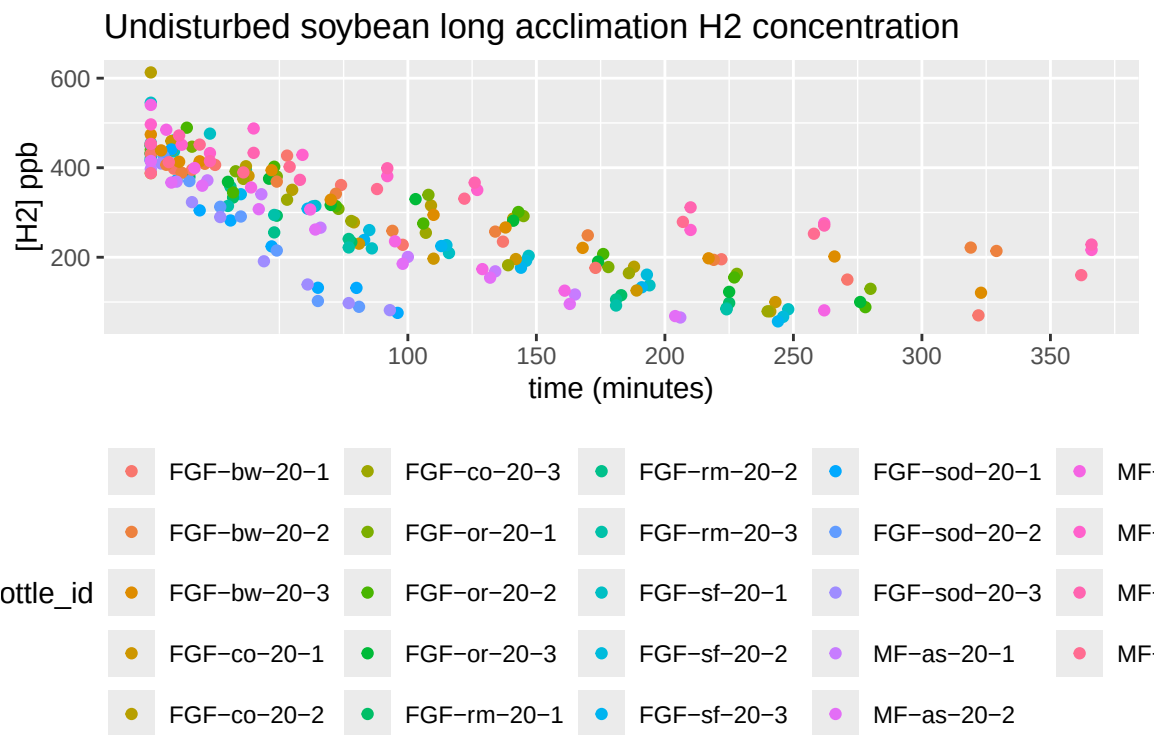
```
## ('geom_point()').
```

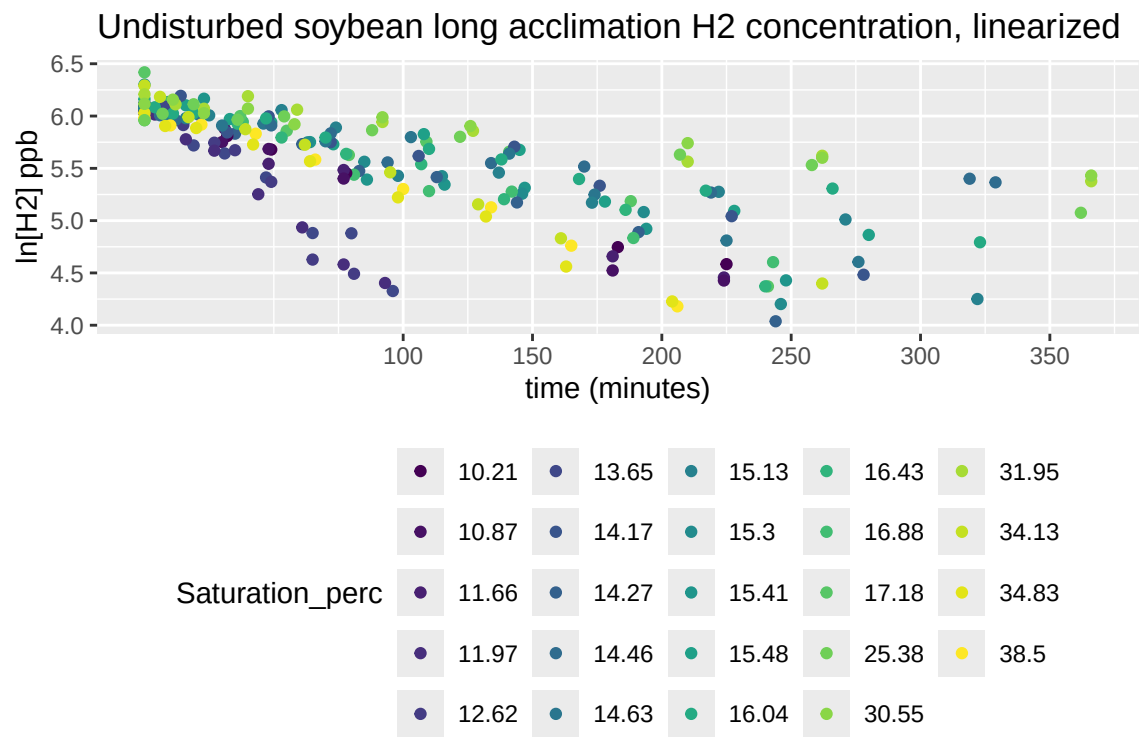


Results

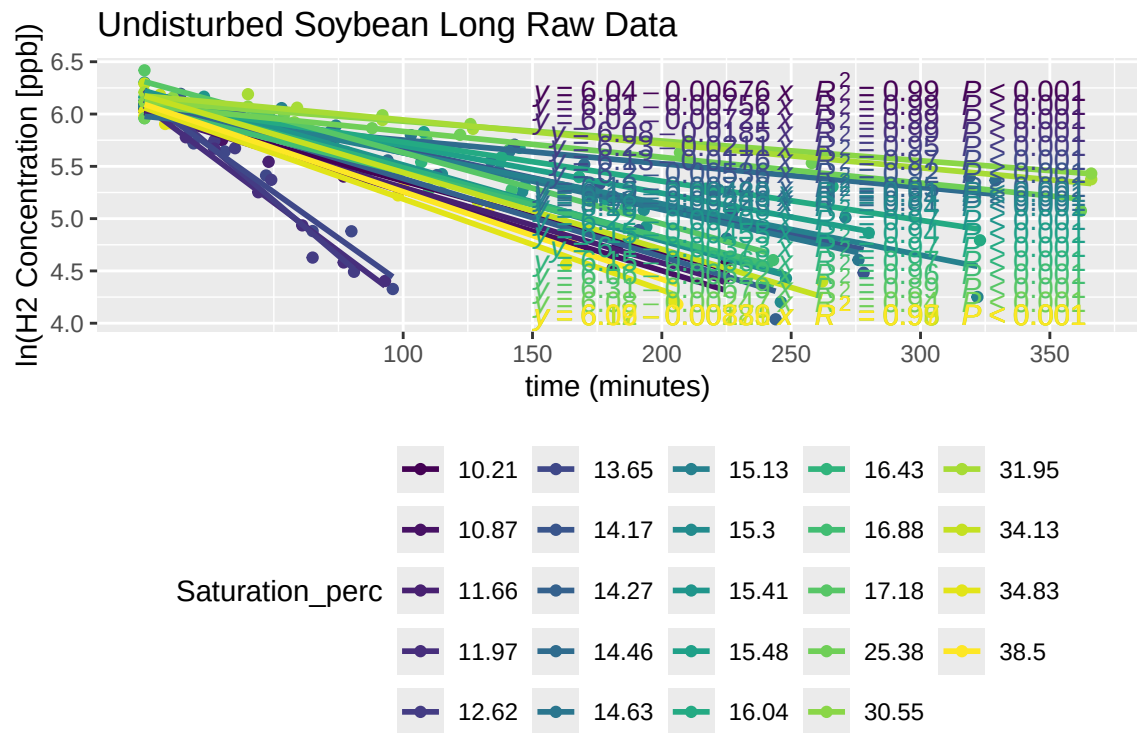
Summer Data

```
## New names:
## * `` -> `...16`
## * `` -> `...18`
## * `` -> `...21`
```

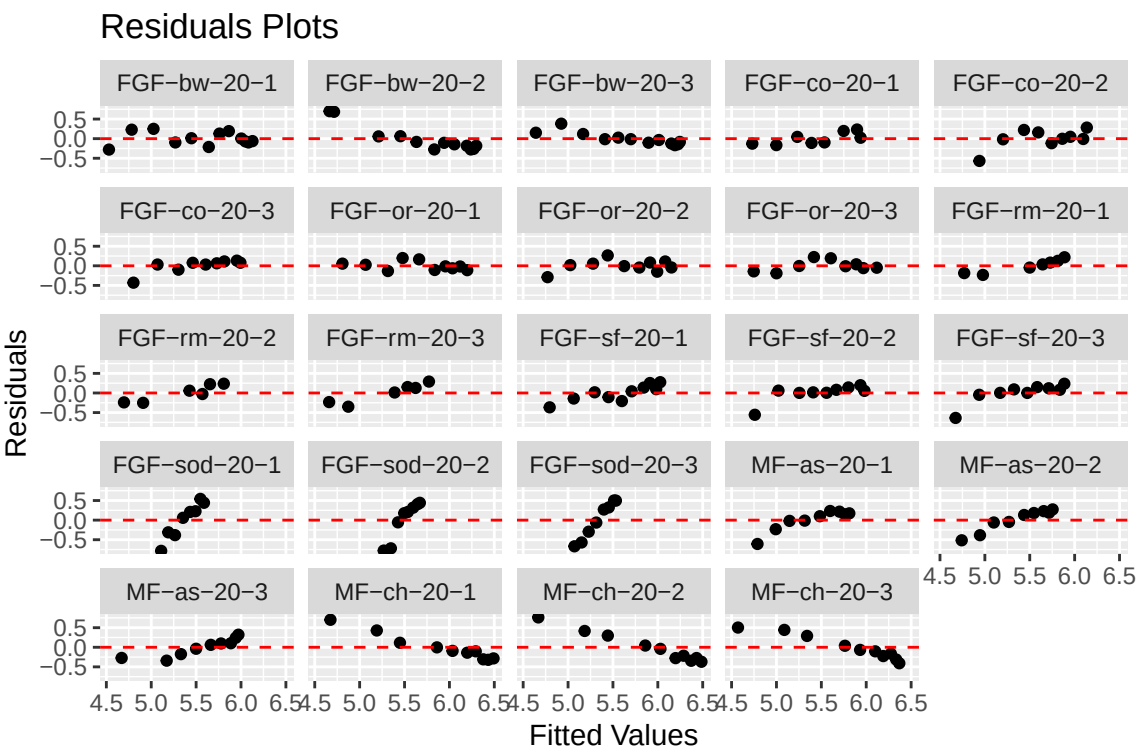




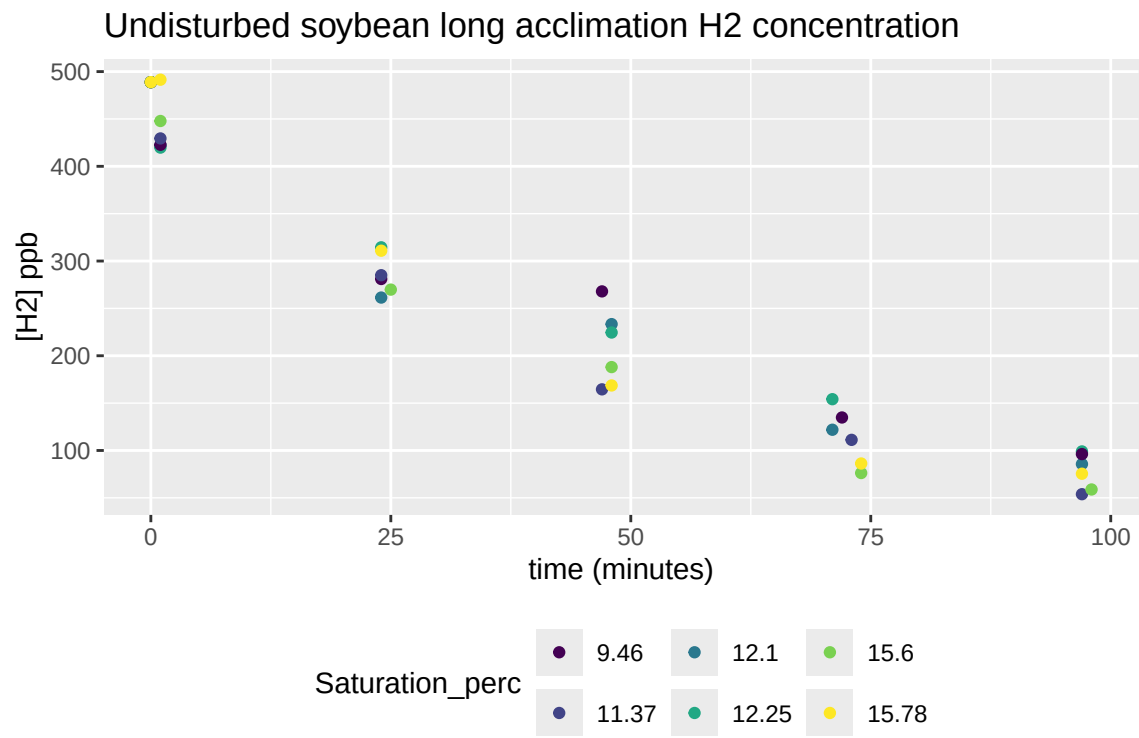
```
## 'geom_smooth()' ` using formula = 'y ~ x'
```

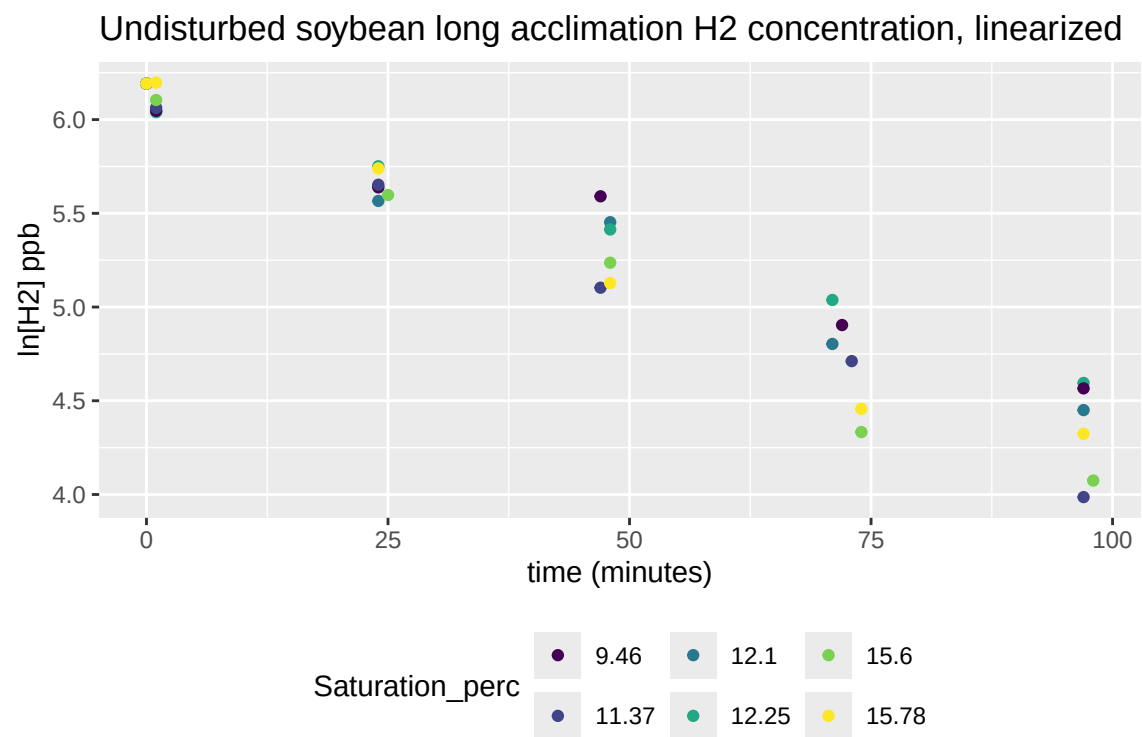


```
## Warning: `fortify(<lm>)` was deprecated in ggplot2 4.0.0.
## i Please use `broom::augment(<lm>)` instead.
## i The deprecated feature was likely used in the ggplot2 package.
## Please report the issue at <https://github.com/tidyverse/ggplot2/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

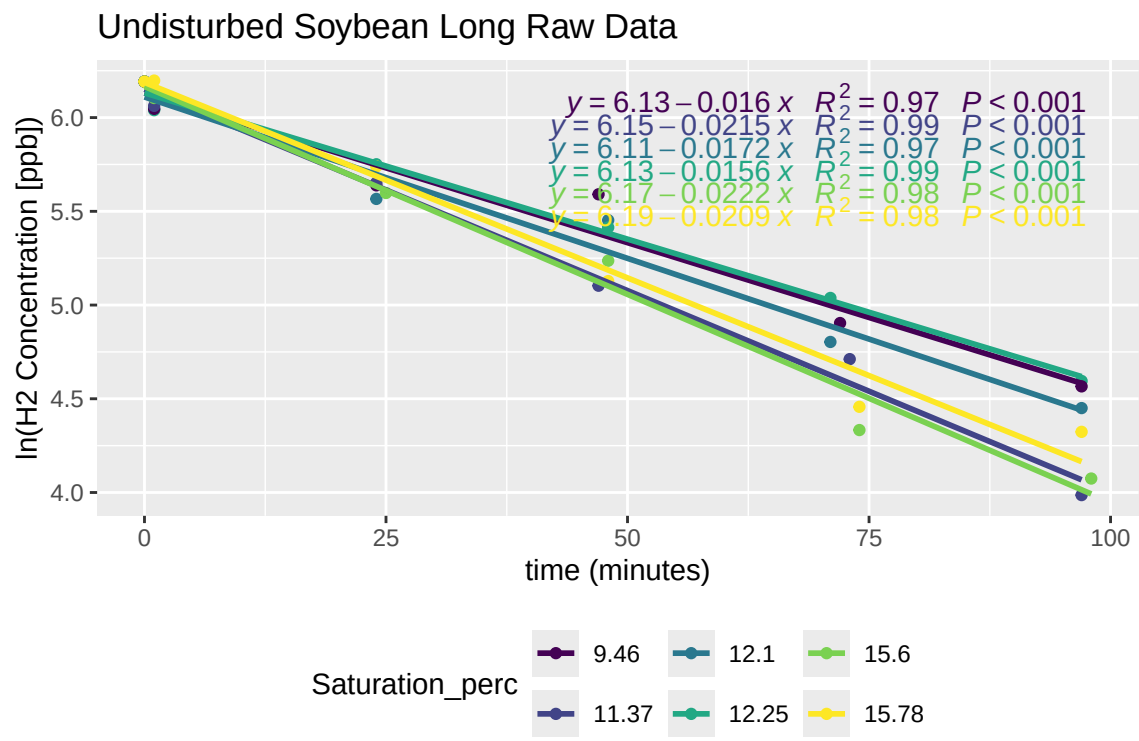


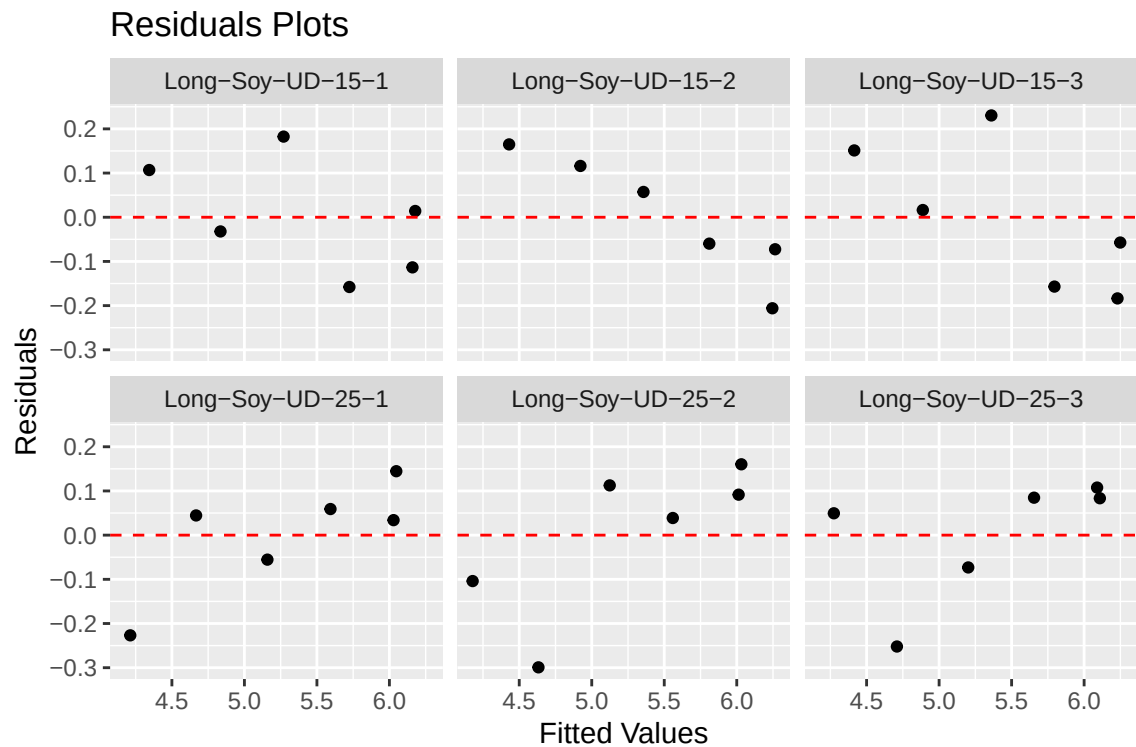
Soy - Undisturbed - Long





```
## 'geom_smooth()' using formula = 'y ~ x'
```





Of note: The fits are really good here, with very small residuals and R^2 values. The linearization seems to be effective without accounting for the asymptotic behavior near the tails, meaning that equilibrium is not sufficiently reached by the end of these incubation experiments within the time frame. I have included the fitting equations in the graph along with the R^2 values - the p values don't mean much because they only compare these lines to a slope of 0, which is obviously out of the question.

```
## Adding missing grouping variables: 'bottle_id'
```

```
## Warning: 'qplot()' was deprecated in ggplot2 3.4.0.
```

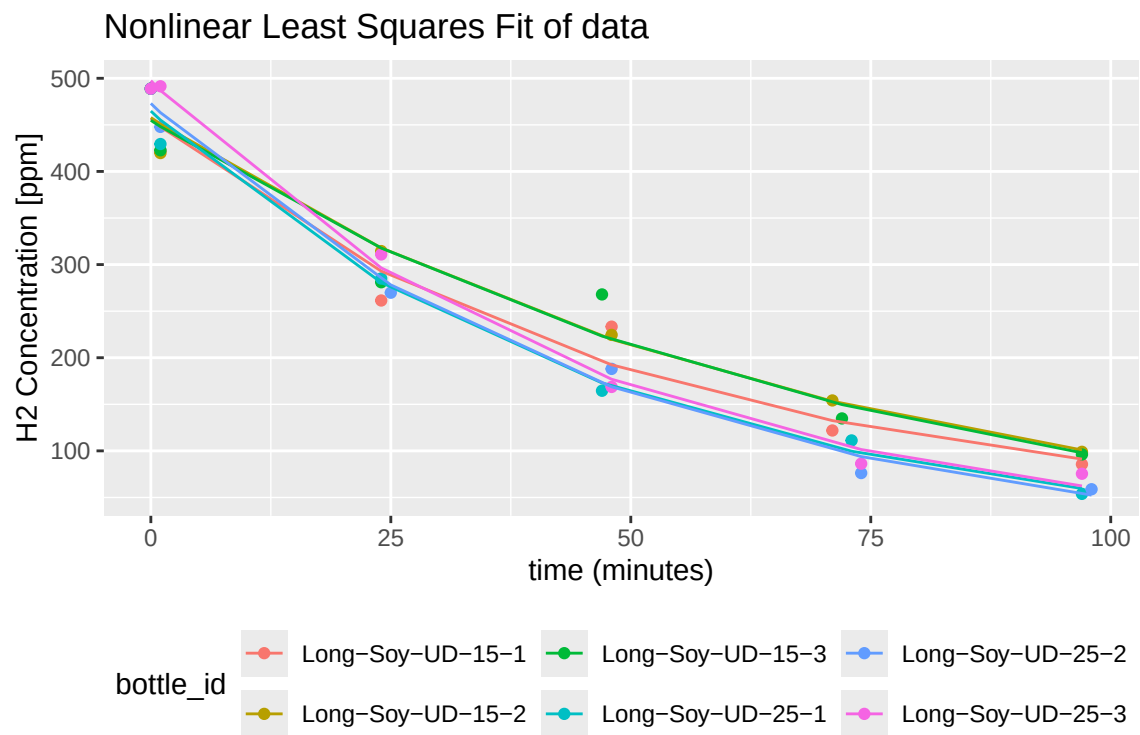
```
## This warning is displayed once every 8 hours.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```



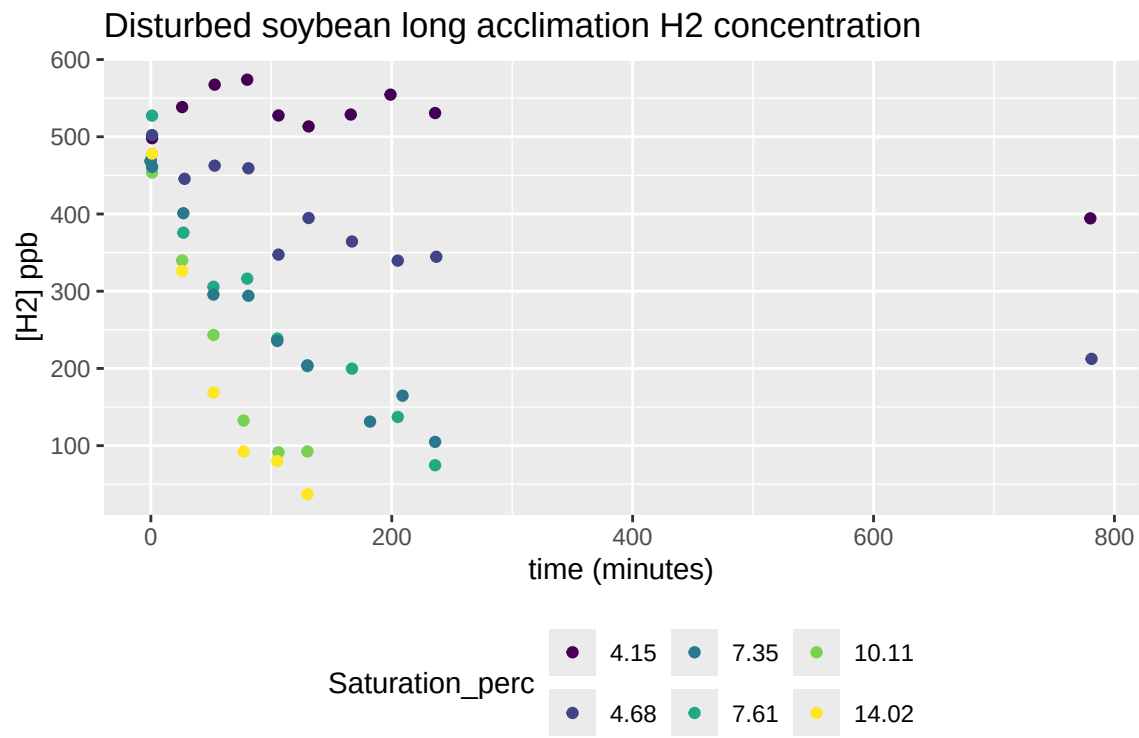
```
## Adding missing grouping variables: 'bottle_id'
```

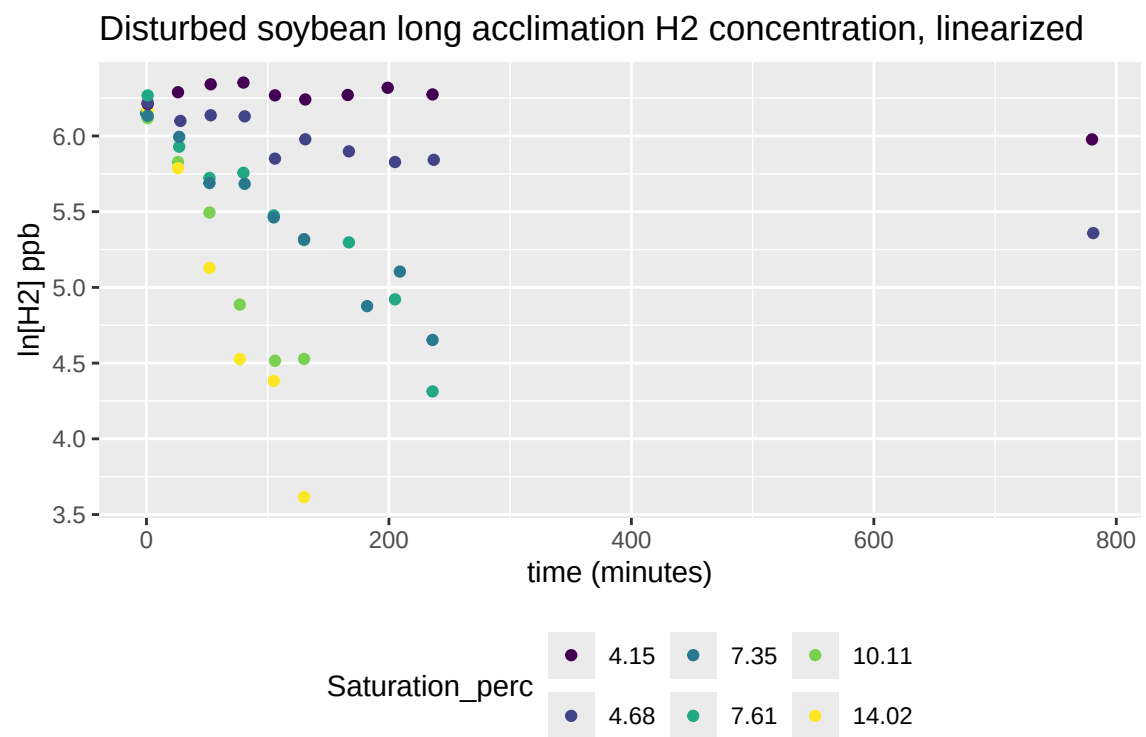


This nonlinear fit is less interpretable than the linearized fits seen above, because, as the data tables show, it is often guessing that the asymptote line is below 0 ppb of H₂, which is impossible. The earlier linearized fitting methods make more physical sense and don't seem to diverge from linearity based upon the residuals plot, but they *may* seem to diverge from homoscedasticity - the residuals plots for each don't seem totally amorphous, but they may be a result of the very small sample sizes per bottle. Regardless, heteroscedasticity does not introduce bias into the model (it doesn't sway the regression coefficient one way or another), but it does mean our models are less precise than the high R² values would make them appear to be.

There are methods that I want to use to seek to address this - the first being a new-to-me method of regression called a robust regression, another being perhaps a background subtraction of the hydrogen background in the nitrogen carrier gas. Most importantly, I think more trial runs will show us how they are behaving at a grander scale and if there are systemic deviations from the assumed linearity of our natural log linearization.

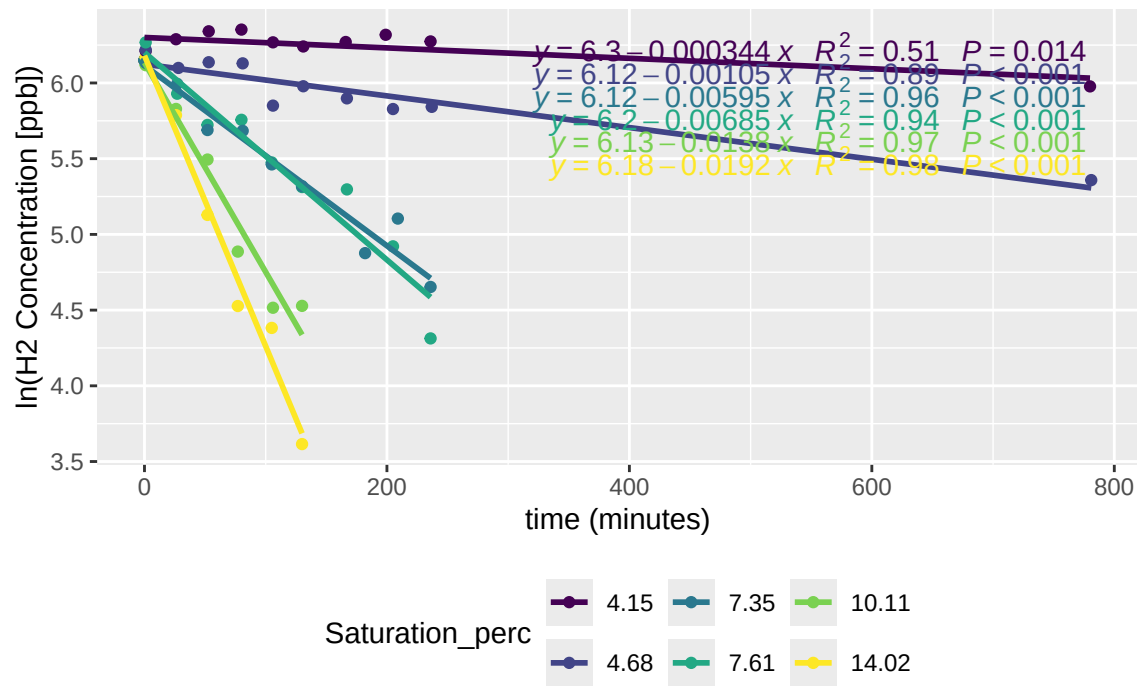
Soy - Disturbed Center - Long

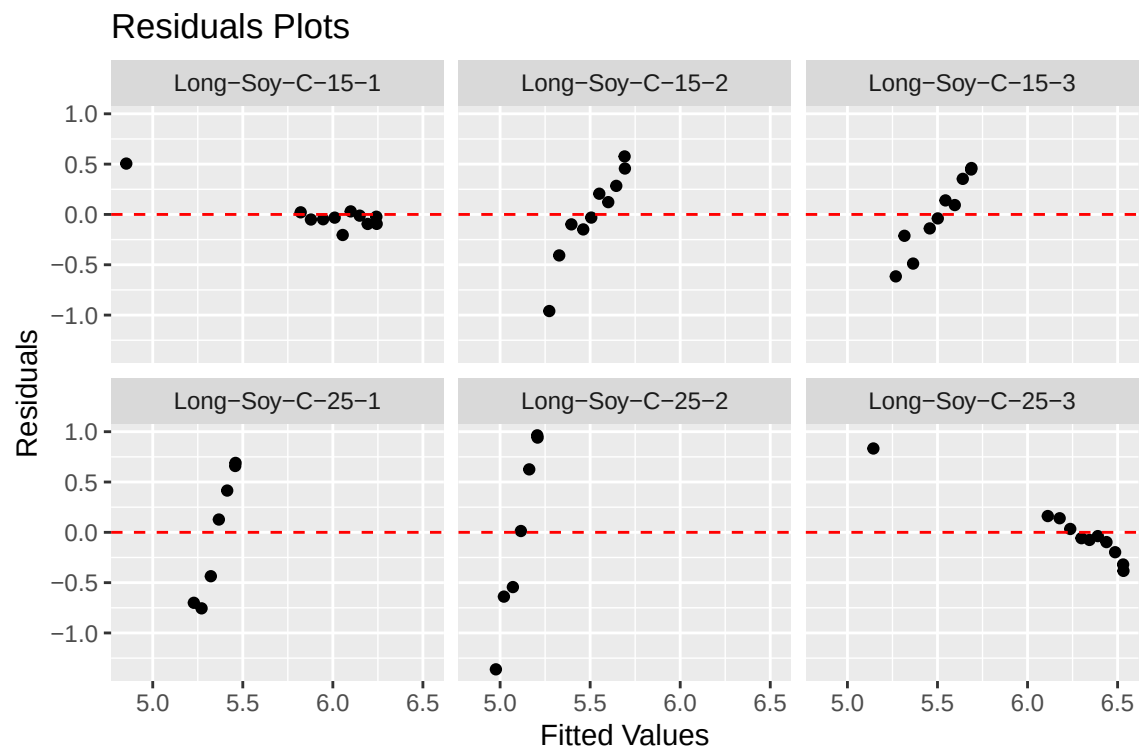




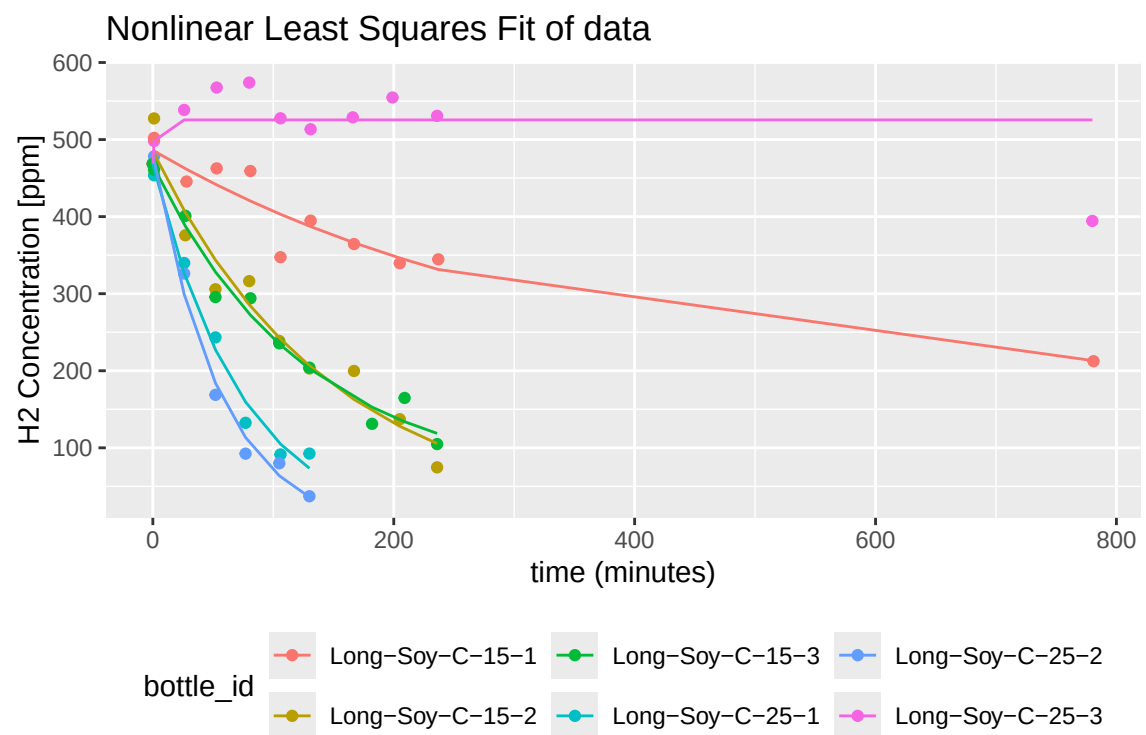
```
## 'geom_smooth()' using formula = 'y ~ x'
```

Example of ln linear fit of first order exponential decay



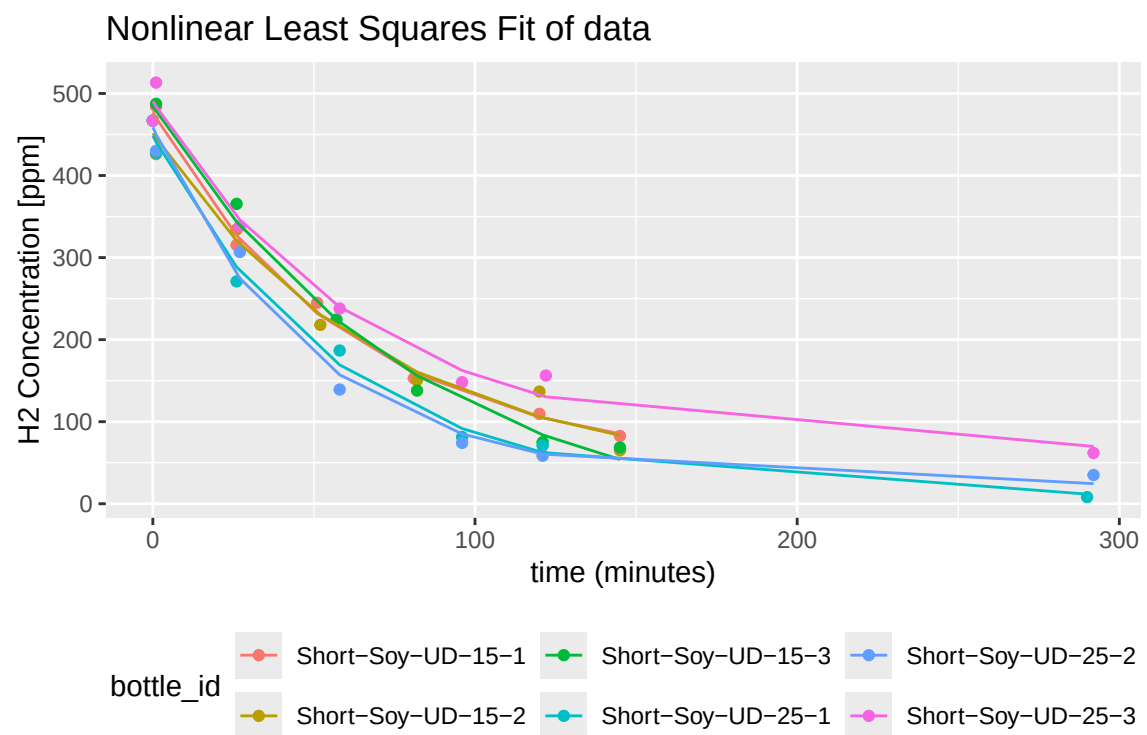


```
## Adding missing grouping variables: 'bottle_id'
```



Good Soy Runs

```
## Adding missing grouping variables: 'bottle_id'
```



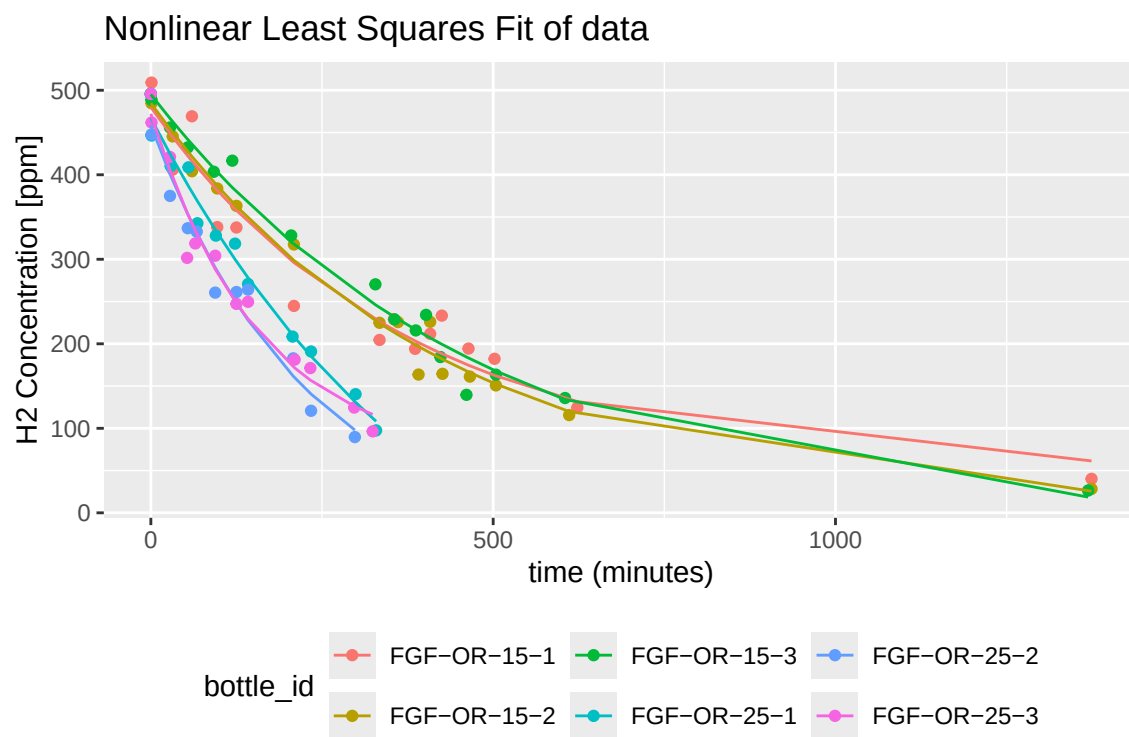
```
## Adding missing grouping variables: 'bottle_id'
```



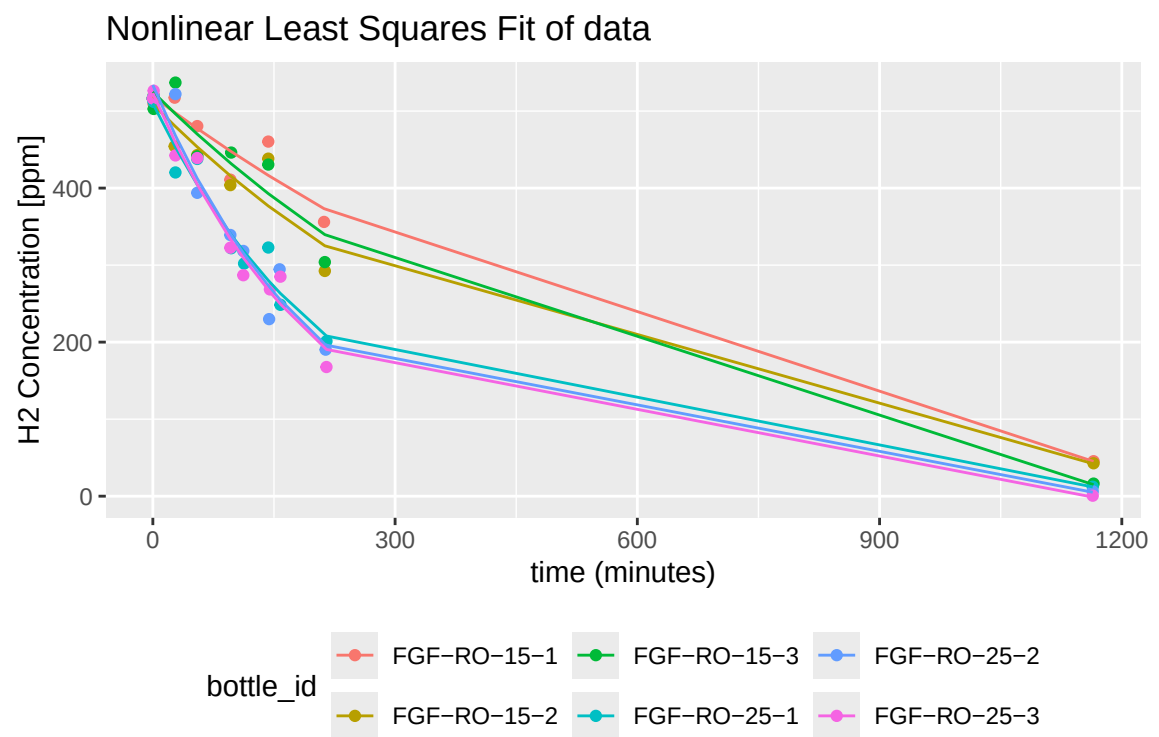
Fairgrown Farms

```
## [1] 77

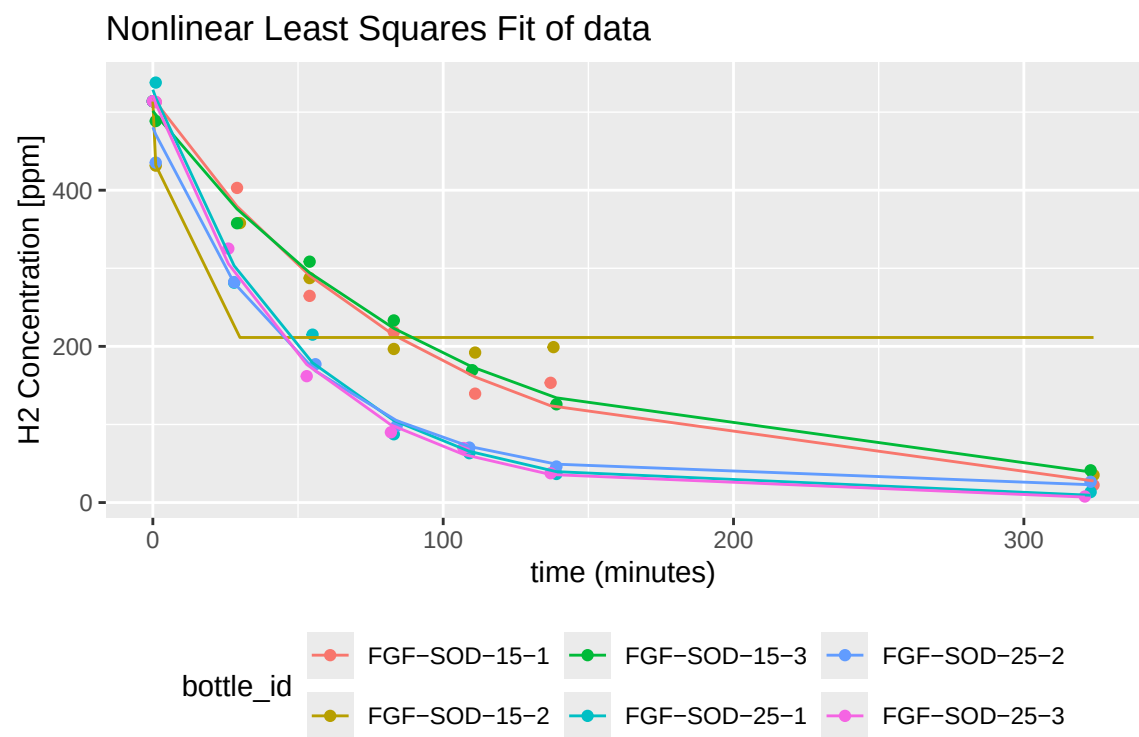
## Adding missing grouping variables: 'bottle_id'
```



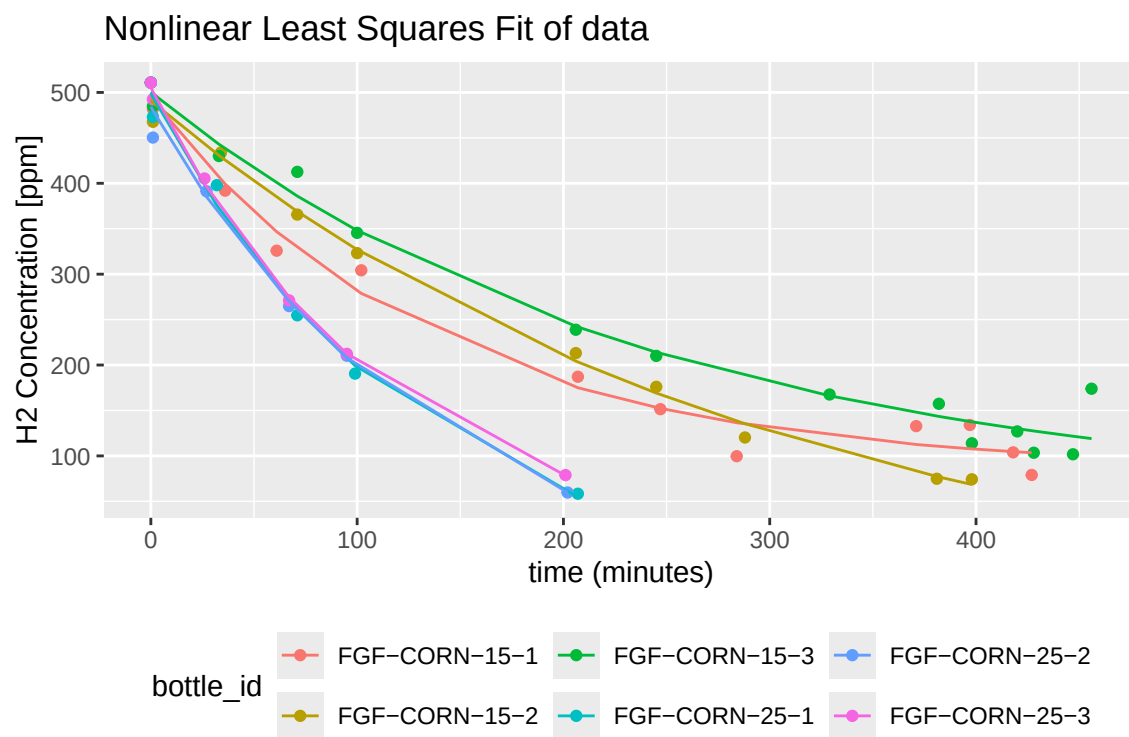
```
## Adding missing grouping variables: 'bottle_id'
```



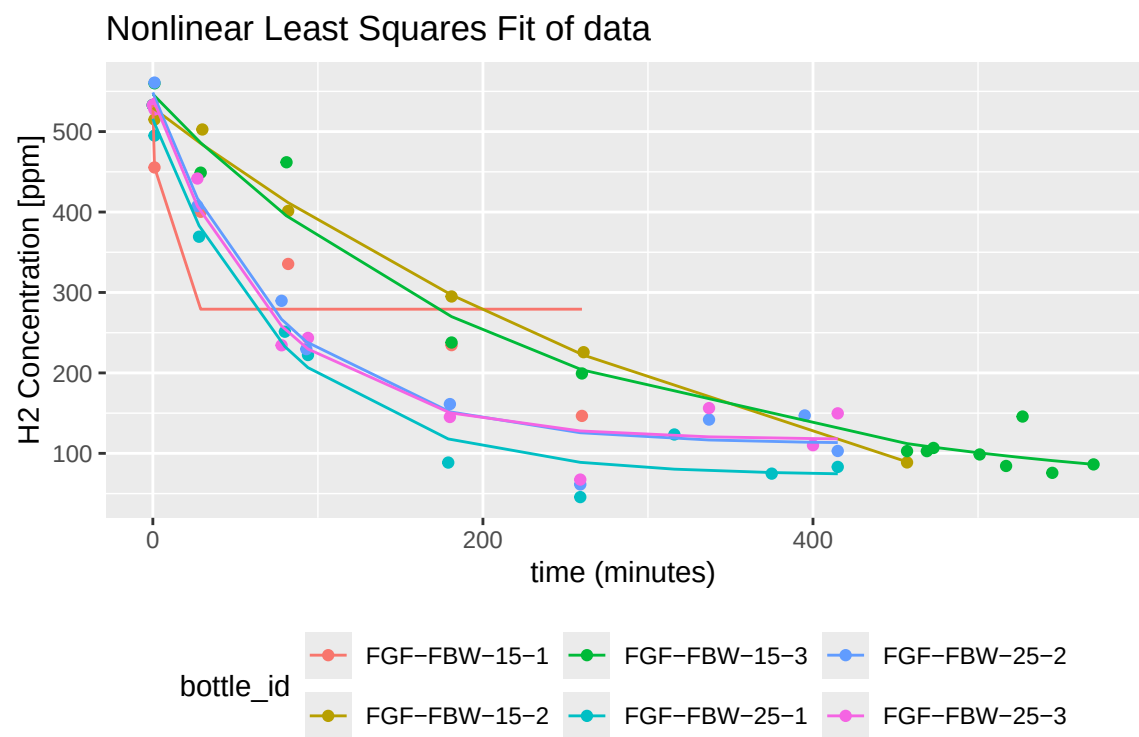
```
## Adding missing grouping variables: 'bottle_id'
```



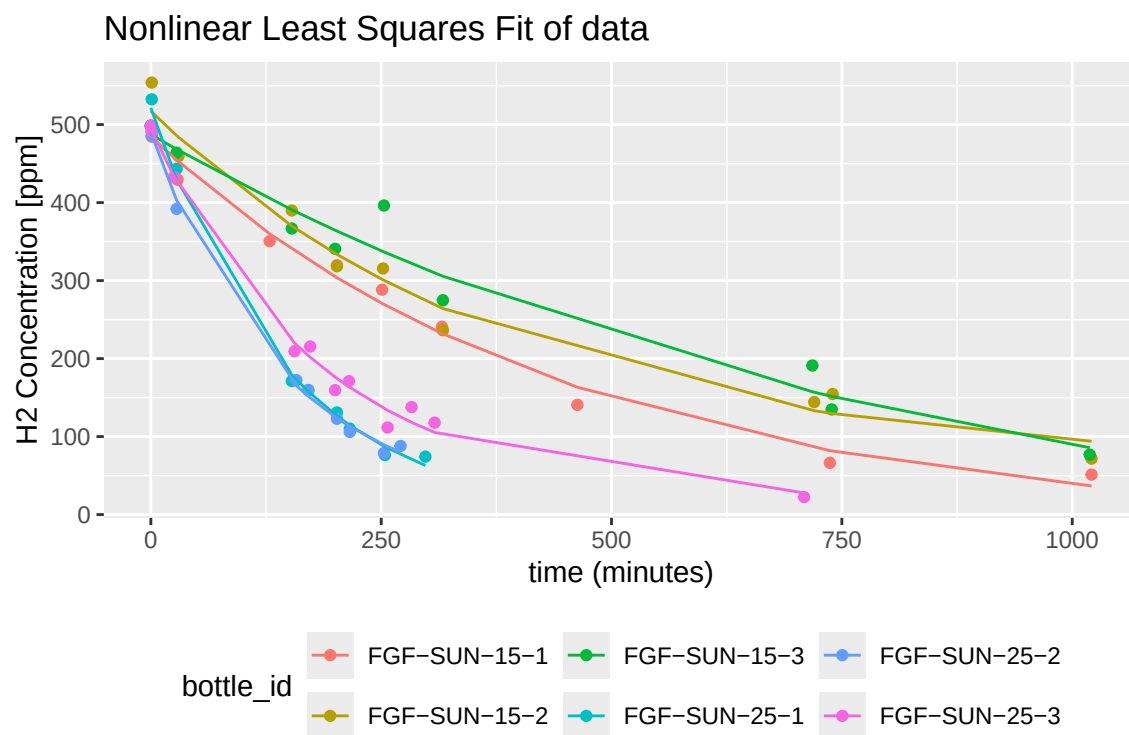
```
## Adding missing grouping variables: 'bottle_id'
```



```
## Adding missing grouping variables: 'bottle_id'
```

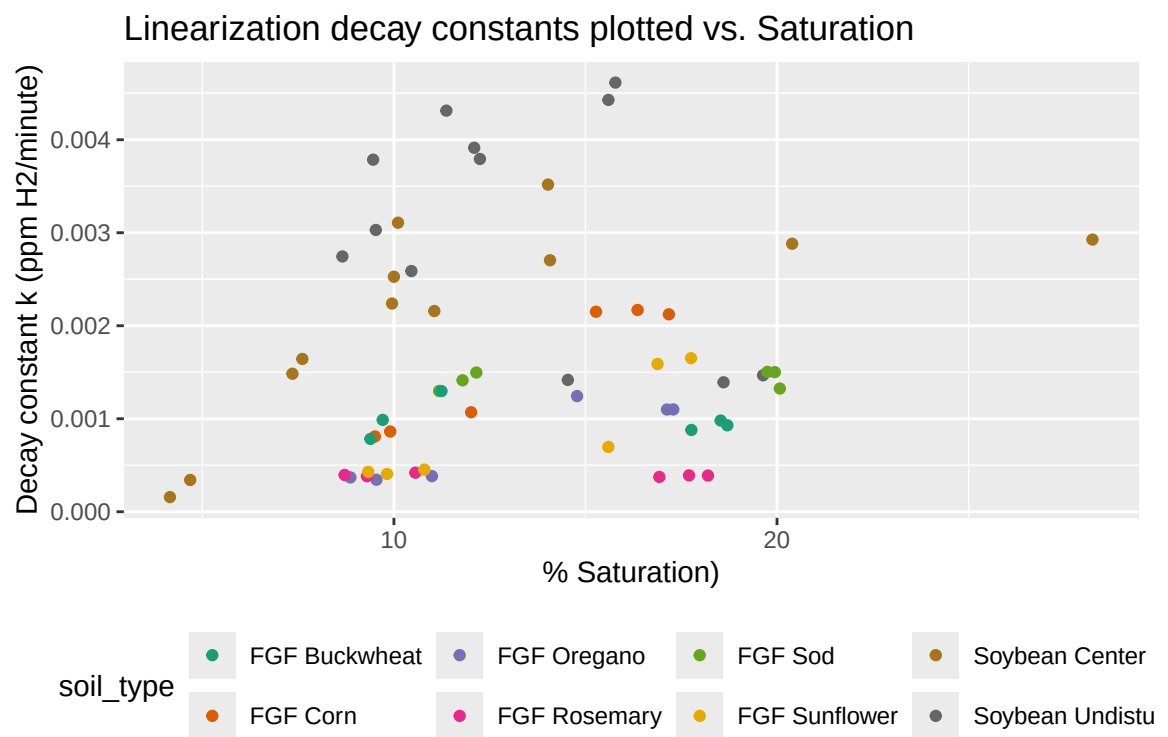


```
## Adding missing grouping variables: 'bottle_id'
```

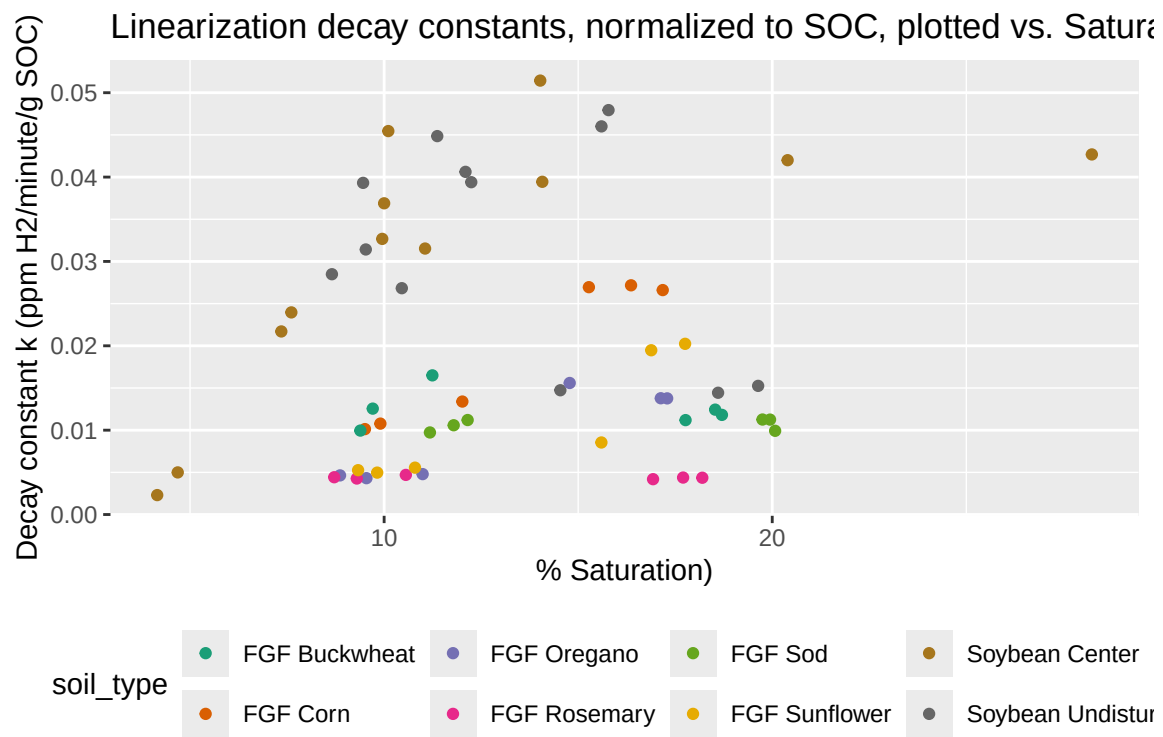


Kinetic Rate Data Summary

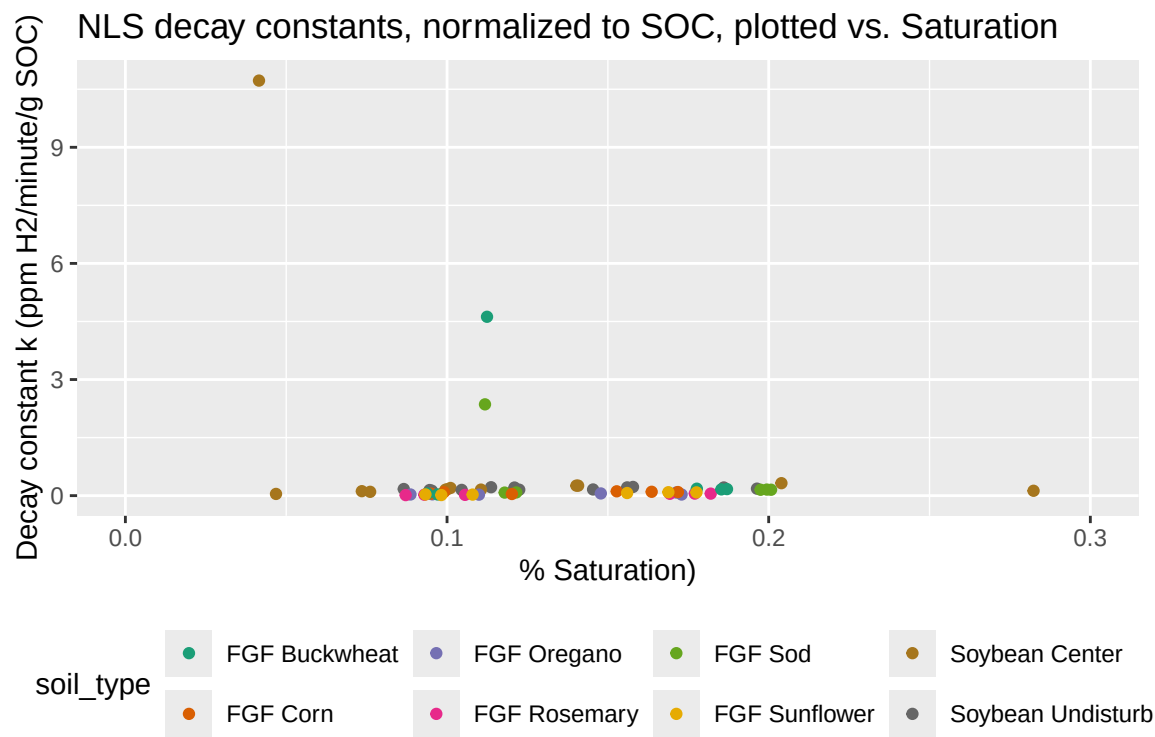
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Include draft work from CEE523 on equation writing.

Discussion

Conclusion

Appendices

Supplementary Information

Code Appendix

code will go here eventually

References

Sand, M., Skeie, R. B., Sandstad, M., Krishnan, S., Myhre, G., Bryant, H., Derwent, R., Hauglustaine, D., Paulot, F., Prather, M., & Stevenson, D., 2023. A multi-model assessment of the Global Warming Potential of hydrogen, *Communications Earth & Environment*, **4**(1), 203.